DATA MODELING AND VISUALIZATION – PRACTICAL VIVA NOTES

Experiments 1 to 7 | BE AI & DS

EXPERIMENT 1: DATA LOADING, STORAGE AND FILE FORMATS

Aim

To load and analyze sales data from multiple file formats such as CSV, Excel, and JSON, and perform data cleaning, transformation, and analysis.

Objective

1. To study different file formats.
2. To load data using Pandas.
3. To perform data cleaning and transformation.
4. To analyze and visualize sales data.

Dataset

Sales data in CSV, Excel, and JSON formats.

Theory

1. Data Loading

Data loading is the process of importing data from different sources into a programming environment for analysis.

Pandas is a Python library commonly used for data loading and manipulation.

2. File Formats

- CSV (Comma-Separated Values): Stores data in rows and columns, separated by commas.
- Excel: Stores data in spreadsheet format with rows, columns, and worksheets.
- JSON (JavaScript Object Notation): Stores data in key-value pairs and supports nested structures.

3. Pandas Functions

- pd.read_csv() – Reads CSV files.
- pd.read_excel() – Reads Excel files.
- pd.read_json() – Reads JSON files.

- df.head() – Displays the first five rows.
- df.info() – Displays dataset information.
- df.describe() – Displays statistical information.
- df.isnull().sum() – Checks missing values.
- df.drop_duplicates() – Removes duplicate records.
- pd.merge() – Merges multiple datasets.

4. Data Cleaning

Data cleaning is the process of identifying and correcting errors, missing values, duplicates, and inconsistencies in a dataset.

5. Data Transformation

Data transformation converts raw data into a suitable format for analysis. It includes changing data types, renaming columns, merging datasets, and creating new columns.

6. Data Integration

Data integration combines data from different sources into one unified dataset.

Practical Steps

1. Import Pandas.
2. Load CSV, Excel, and JSON files.
3. Explore the datasets using head(), info(), and describe().
4. Check missing values.
5. Remove duplicates.
6. Clean and transform data.
7. Merge datasets if required.
8. Perform statistical analysis.
9. Create visualizations.

Conclusion

In this experiment, we successfully loaded and analyzed sales data from different file formats, including CSV, Excel, and JSON. We performed data cleaning, transformation, and analysis using Pandas.

Viva Questions and Answers

Q1. What is the difference between CSV, Excel, and JSON?

Ans: CSV stores tabular data separated by commas. Excel stores data in spreadsheets with multiple sheets. JSON stores data in key-value pairs and supports nested data.

Q2. Which Python library is used to load datasets?

Ans: Pandas.

Q3. How do you load a CSV file?

Ans: Using pd.read_csv('filename.csv').

Q4. How do you load an Excel file?

Ans: Using pd.read_excel('filename.xlsx').

Q5. How do you load a JSON file?

Ans: Using pd.read_json('filename.json').

Q6. How do you check missing values?

Ans: Using df.isnull().sum() or df.info().

Q7. How do you handle missing values?

Ans: Using fillna(), dropna(), or interpolation.

Q8. How do you remove duplicate records?

Ans: Using df.drop_duplicates().

Q9. How do you merge two datasets?

Ans: Using pd.merge(), pd.concat(), or join().

Q10. What is data transformation?

Ans: Data transformation is the process of converting raw data into a suitable format for analysis.

Q11. What is a DataFrame?

Ans: A DataFrame is a two-dimensional tabular data structure in Pandas containing rows and columns.

Q12. What is the difference between structured and unstructured data?

Ans: Structured data is organized in rows and columns, while unstructured data does not follow a fixed tabular structure, such as images, videos, and text documents.

Q13. What is data migration?

Ans: Data migration is the process of transferring data from one storage system or format to another.

Q14. Why do we convert data into a unified format?

Ans: To maintain consistency and make data processing and analysis easier.

EXPERIMENT 2: INTERACTING WITH WEB APIs

Aim

To interact with the OpenWeatherMap API and analyze weather data for a specific location.

Objective

1. To understand web APIs.
2. To retrieve weather data using an API.
3. To extract temperature, humidity, and wind speed.
4. To clean, analyze, and visualize weather data.

Dataset

Weather data retrieved from the OpenWeatherMap API.

Theory

1. What is an API?

API stands for Application Programming Interface. It allows two software applications to communicate and exchange data.

2. OpenWeatherMap API

OpenWeatherMap is a web service that provides weather information such as temperature, humidity, wind speed, and atmospheric pressure.

It provides:

- Current weather data.
- Forecast data.
- Historical weather data.

3. API Key

An API key is a unique authentication key used to access an API. It helps identify users and manage access.

4. HTTP Request

An HTTP request is sent to a server to retrieve or exchange information.

5. JSON

JSON stands for JavaScript Object Notation. It is a lightweight format used to store and exchange data.

6. Requests Library

Python's requests library is used to send HTTP requests to web APIs.

7. Weather Parameters

- Temperature: Measures how hot or cold the atmosphere is.
- Humidity: Amount of moisture in the air.
- Pressure: Atmospheric pressure.
- Wind speed: Speed of moving air.
- Precipitation: Rainfall or snowfall.

Practical Steps

1. Register on OpenWeatherMap.
2. Generate an API key.
3. Import the Requests and Pandas libraries.
4. Send an HTTP request.
5. Receive weather data in JSON format.
6. Extract temperature, humidity, and wind speed.
7. Clean and preprocess the data.
8. Perform statistical analysis.
9. Visualize the results using graphs.

Conclusion

In this experiment, we retrieved weather data using the OpenWeatherMap API, extracted important weather parameters, and analyzed and visualized weather patterns.

Viva Questions and Answers

Q1. What is an API?

Ans: API stands for Application Programming Interface. It allows two applications to communicate with each other.

Q2. Why do we use APIs in data analysis?

Ans: APIs help retrieve data automatically from external sources.

Q3. What is OpenWeatherMap API?

Ans: It is a weather service API that provides current, forecast, and historical weather information.

Q4. What is an API key?

Ans: An API key is a unique authentication key used to access an API.

Q5. Which Python library is used to send API requests?

Ans: The requests library.

Q6. What is the typical response format of OpenWeatherMap?

Ans: JSON format.

Q7. What is JSON?

Ans: JSON stands for JavaScript Object Notation. It stores data in key-value pairs.

Q8. How do you extract temperature from a JSON response?

Ans: Using response['main']['temp'].

Q9. What is HTTP?

Ans: HTTP stands for Hypertext Transfer Protocol. It is used for communication between clients and web servers.

Q10. What is the difference between GET and POST?

Ans: GET is used to retrieve data, while POST is used to send data to a server.

Q11. What are the important weather parameters?

Ans: Temperature, humidity, pressure, wind speed, and precipitation.

Q12. Which graphs are suitable for weather data?

Ans: Line charts, bar charts, scatter plots, and heatmaps.

Q13. Why is data preprocessing required for API data?

Ans: To handle missing values, correct data types, and inconsistent formats.

Q14. What are the advantages of APIs?

Ans: Real-time data retrieval, automation, scalability, and easy data updating.

Q15. What is the use of latitude and longitude?

Ans: They identify the geographical location of a place.

EXPERIMENT 3: DATA CLEANING AND PREPARATION

Aim

To perform data cleaning and preparation to gain insights into the factors contributing to customer churn.

Objective

1. To identify missing values.
2. To remove duplicate records.
3. To correct inconsistent data.
4. To handle outliers.
5. To prepare data for analysis and machine learning.

Dataset

Telecom_Customer_Churn.csv

Theory

1. Data Cleaning

Data cleaning is the process of identifying and correcting errors, missing values, duplicates, and inconsistencies in a dataset.

2. Customer Churn

Customer churn occurs when customers stop using a company's products or services.

For example, when a customer stops using a telecom company's network.

3. Missing Values

Missing values are empty or unavailable values in a dataset.

Pandas functions:

- df.isnull() – Identifies missing values.

- df.isnull().sum() – Counts missing values column-wise.
- df.dropna() – Removes rows or columns containing missing values.
- df.fillna() – Replaces missing values.

4. Duplicate Records

Duplicate records are repeated entries in a dataset.

- df.duplicated() – Identifies duplicate records.
- df.drop_duplicates() – Removes duplicate records.

5. Outliers

Outliers are data points that are significantly different from other observations.

Methods to detect outliers:

- Box plot.
- IQR method.
- Z-score method.

6. Feature Engineering

Feature engineering is the process of creating new features from existing data to improve analysis or model performance.

Example: Creating a tenure group from customer tenure.

7. Normalization

Normalization is the process of bringing numerical values to a common scale.

It is useful for distance-based machine learning algorithms.

8. Data Splitting

Data is divided into training and testing sets.

- Training data: Used to train the model.
- Testing data: Used to evaluate the model.

Practical Steps

1. Import Pandas and other required libraries.
2. Load the telecom customer churn dataset.
3. Explore the dataset.

4. Identify missing values.
5. Handle missing values.
6. Remove duplicates.
7. Correct inconsistent data.
8. Handle outliers.
9. Perform feature engineering.
10. Normalize or scale data if required.
11. Split data into training and testing sets.

Conclusion

In this experiment, we explored data cleaning and preparation techniques and handled missing values, duplicates, inconsistencies, and outliers in the Telecom Customer Churn dataset.

Viva Questions and Answers

Q1. What is customer churn?

Ans: Customer churn occurs when customers stop using a company's products or services.

Q2. Why is customer churn analysis important?

Ans: It helps companies understand why customers leave and develop strategies to retain them.

Q3. How do you identify missing values?

Ans: Using df.isnull().sum().

Q4. How can you handle missing values?

Ans: By removing rows or filling missing values using mean, median, mode, or interpolation.

Q5. How do you detect duplicate records?

Ans: Using df.duplicated().

Q6. How do you remove duplicate records?

Ans: Using df.drop_duplicates().

Q7. What are data inconsistencies?

Ans: Errors such as different spellings, inconsistent capitalization, and incorrect formats.

Q8. Why do we convert columns into correct data types?

Ans: To ensure accurate calculations and analysis.

Q9. Which function changes the data type in Pandas?

Ans: astype().

Q10. What are outliers?

Ans: Outliers are values that differ significantly from the rest of the dataset.

Q11. How can you detect outliers?

Ans: Using box plots, IQR, and Z-score methods.

Q12. What is feature engineering?

Ans: Creating new meaningful features from existing data.

Q13. What is normalization?

Ans: Normalization brings numerical features to a common scale.

Q14. Why is normalization important?

Ans: It prevents features with larger numerical values from dominating distance-based algorithms.

Q15. What is train-test splitting?

Ans: It is the process of dividing data into training and testing sets.

Q16. Which function is used for train-test splitting?

Ans: train_test_split() from Scikit-learn.

EXPERIMENT 4: DATA WRANGLING ON REAL ESTATE MARKET

Aim

To perform data wrangling on real estate market data.

Objective

To clean, transform, organize, and prepare real estate data for further analysis or modeling.

Dataset

RealEstate_Prices.csv

Theory

1. Data Wrangling

Data wrangling, also known as data munging, is the process of cleaning, transforming, and organizing raw data into a suitable format for analysis.

2. Data Integration

Data integration combines data from multiple sources into a unified dataset.

3. Data Cleaning

It involves handling missing values, duplicates, incorrect values, and inconsistent formats.

4. Data Transformation

It involves changing data formats, renaming columns, encoding categorical variables, and creating new features.

5. Categorical Variables

Categorical variables contain labels or categories rather than numerical measurements.

Example: Property type, city, and neighborhood.

6. Encoding

Encoding converts categorical data into numerical form.

- Label Encoding: Assigns a numerical value to each category.
- One-Hot Encoding: Creates separate binary columns for each category.

7. Data Aggregation

It summarizes data using operations such as sum, mean, median, and count.

Example: Calculating average property prices by neighborhood.

8. Outlier Handling

Outliers are extreme values that may affect analysis or model performance. They can be detected using box plots, IQR, or Z-score.

Practical Steps

1. Import Pandas and required libraries.

2. Load RealEstate_Prices.csv.
3. Explore the dataset.
4. Clean column names.
5. Handle missing values.
6. Remove duplicates.
7. Merge datasets if required.
8. Filter and subset relevant data.
9. Encode categorical variables.
10. Calculate average sale prices by neighborhood.
11. Detect and handle outliers.

Conclusion

In this experiment, we performed data preprocessing and wrangling techniques on the Real Estate Prices dataset to prepare it for further analysis and modeling.

Viva Questions and Answers

Q1. What is data wrangling?

Ans: Data wrangling is the process of cleaning, transforming, and organizing raw data for analysis.

Q2. Why is data wrangling important?

Ans: It improves data quality, consistency, and accuracy.

Q3. What is the difference between data cleaning and data transformation?

Ans: Data cleaning corrects errors and missing values, while data transformation changes data into a suitable format.

Q4. How do you clean column names?

Ans: Using string functions such as:

df.columns = df.columns.str.replace(' ', '_').str.lower()

Q5. How can you handle missing values?

Ans: Using dropna(), fillna(), or interpolation.

Q6. How do you merge datasets in Pandas?

Ans: Using pd.merge().

Q7. What is filtering?

Ans: Filtering selects only the records that satisfy a particular condition.

Q8. What is subsetting?

Ans: Subsetting means selecting specific rows or columns from a dataset.

Q9. What is label encoding?

Ans: Label encoding assigns a numerical value to each category.

Q10. What is one-hot encoding?

Ans: One-hot encoding creates separate binary columns for each category.

Q11. Difference between label encoding and one-hot encoding?

Ans: Label encoding assigns numbers to categories, while one-hot encoding creates separate columns containing 0 and 1.

Q12. How do you calculate average sale price by neighborhood?

Ans: Using:

df.groupby('Neighborhood')['SalePrice'].mean()

Q13. Why are outliers important in real estate data?

Ans: Outliers can affect average prices, distort analysis, and reduce model accuracy.

Q14. Which methods are used to detect outliers?

Ans: Box plot, IQR, and Z-score.

Q15. What are the next steps after data wrangling?

Ans: Data visualization, exploratory data analysis, feature selection, and model building.

EXPERIMENT 5: DATA VISUALIZATION USING MATPLOTLIB

Aim

To perform data visualization using Matplotlib for analyzing Air Quality Index trends in a city.

Objective

1. To understand data visualization.
2. To use Matplotlib for creating graphs.

3. To visualize AQI and pollutant trends.
4. To identify patterns and relationships in air quality data.

Dataset

City_Air_Quality.csv

Theory

1. Data Visualization

Data visualization is the graphical representation of data using charts, graphs, and plots.

It helps identify patterns, trends, relationships, and outliers.

2. Matplotlib

Matplotlib is a Python library used to create static, animated, and interactive visualizations.

3. Pyplot

Pyplot is a module of Matplotlib that provides functions for creating graphs.

Import statement:

import matplotlib.pyplot as plt

4. Types of Charts

- Line Chart: Shows trends over time.
- Bar Chart: Compares different categories.
- Histogram: Shows frequency distribution.
- Pie Chart: Shows percentage contribution.
- Scatter Plot: Shows relationships between two numerical variables.
- Box Plot: Shows distribution, median, quartiles, and outliers.

5. AQI

AQI stands for Air Quality Index. It indicates the quality of air and the level of pollution.

6. Pollutants

- PM2.5: Fine particulate matter with a diameter of 2.5 micrometers or less.
- PM10: Particulate matter with a diameter of 10 micrometers or less.
- CO: Carbon Monoxide.

7. Plot Customization

Plots can be customized using titles, axis labels, legends, colors, and gridlines.

Important Matplotlib Functions

- plt.plot() – Line plot.
- plt.bar() – Bar chart.
- plt.hist() – Histogram.
- plt.pie() – Pie chart.
- plt.scatter() – Scatter plot.
- plt.boxplot() – Box plot.
- plt.title() – Adds title.
- plt.xlabel() – Adds X-axis label.
- plt.ylabel() – Adds Y-axis label.
- plt.legend() – Displays legend.
- plt.grid() – Displays grid.
- plt.show() – Displays the graph.

Practical Steps

1. Import Pandas and Matplotlib.
2. Load the air quality dataset.
3. Explore the dataset.
4. Identify date, AQI, and pollutant columns.
5. Create line plots for AQI trends.
6. Create bar plots for comparisons.
7. Create box plots to analyze distributions.
8. Create scatter plots to identify relationships.
9. Add titles, labels, and legends.
10. Display the graphs.

Conclusion

In this experiment, we used Matplotlib to visualize AQI trends and pollutant levels. We created different types of graphs to identify patterns and relationships in air quality data.

Viva Questions and Answers

Q1. What is Matplotlib?

Ans: Matplotlib is a Python library used for creating data visualizations.

Q2. How do you import Matplotlib?

Ans: import matplotlib.pyplot as plt

Q3. Which plot is best for showing AQI trends over time?

Ans: Line plot.

Q4. How do you plot multiple pollutants on the same graph?

Ans: By using multiple plt.plot() statements before calling plt.show().

Q5. What is the purpose of a bar chart?

Ans: It compares values across different categories.

Q6. What information does a box plot provide?

Ans: It shows the median, quartiles, data distribution, and outliers.

Q7. How do you add a title to a graph?

Ans: Using plt.title().

Q8. How do you add X-axis and Y-axis labels?

Ans: Using plt.xlabel() and plt.ylabel().

Q9. What is the use of a legend?

Ans: A legend identifies different data series shown in a graph.

Q10. What is a scatter plot used for?

Ans: It is used to identify relationships between two numerical variables.

Q11. How do you change the size of a graph?

Ans: Using plt.figure(figsize=(width, height)).

Q12. How do you display a graph?

Ans: Using plt.show().

Q13. What is the difference between a line chart and a bar chart?

Ans: A line chart shows trends over time, while a bar chart compares categories.

Q14. What is the difference between Pyplot and Object-Oriented interfaces?

Ans: Pyplot uses simple functions such as plt.plot(). The Object-Oriented interface uses Figure and Axes objects and provides more control over complex plots.

Q15. Why is data visualization important?

Ans: It makes complex data easier to understand and helps identify trends, patterns, and anomalies.

Q16. What are interactive visualizations?

Ans: Interactive visualizations allow users to interact with graphs using controls such as sliders and dynamic updates.

EXPERIMENT 6: DATA AGGREGATION

Aim

To analyze and visualize retail sales data by aggregating and comparing sales across regions and product categories.

Objective

1. To understand data aggregation.
2. To group sales data by region.
3. To calculate total sales.
4. To identify top-performing regions.
5. To visualize sales distribution.

Dataset

Retail_Sales_Data.csv

Theory

1. Data Aggregation

Data aggregation is the process of combining data from multiple sources or summarizing data from a single source to produce meaningful information.

It helps identify trends, patterns, and relationships.

2. Common Aggregation Operations

- Sum: Calculates the total.
- Mean: Calculates the average.
- Median: Finds the middle value.
- Mode: Finds the most frequently occurring value.
- Count: Counts the number of non-null values.

3. Grouping

Grouping organizes data into categories based on a particular column.

Pandas function:

df.groupby()

4. Aggregation vs Grouping

Grouping organizes records into categories, while aggregation calculates summary values such as sum, mean, and count for those groups.

5. Stacked Bar Plot

A stacked bar plot displays multiple categories as parts of a total bar. It helps compare total sales and category contributions across regions.

6. Benefits of Data Aggregation

- Improved efficiency.
- Increased accuracy.
- Better decision-making.
- Identification of top-performing regions.
- Easier analysis of large datasets.

Important Pandas Functions

- df.groupby() – Groups data.
- sum() – Calculates total.
- mean() – Calculates average.
- median() – Calculates median.
- mode() – Calculates mode.
- count() – Counts records.
- max() – Finds maximum.

- min() – Finds minimum.
- sort_values() – Sorts values.

Practical Steps

1. Import Pandas and Matplotlib.
2. Load Retail_Sales_Data.csv.
3. Explore the dataset.
4. Identify region, sales amount, and product category.
5. Group sales by region.
6. Calculate total sales for each region.
7. Create bar plots or pie charts.
8. Identify the top-performing regions.
9. Group data by region and product category.
10. Create grouped or stacked bar plots.

Conclusion

In this experiment, we performed data aggregation on retail sales data. We calculated total sales by region and product category, visualized sales distribution, and identified top-performing regions.

Viva Questions and Answers

Q1. What is data aggregation?

Ans: Data aggregation is the process of summarizing raw data to obtain meaningful information such as totals, averages, and counts.

Q2. Which Pandas function is used for data aggregation?

Ans: groupby().

Q3. How do you calculate total sales by region?

Ans:

df.groupby('Region')['SalesAmount'].sum()

Q4. What is the difference between grouping and aggregation?

Ans: Grouping organizes data into categories, while aggregation calculates summary values for those categories.

Q5. How do you find the top-performing region?

Ans: By sorting aggregated sales in descending order using sort_values(ascending=False).

Q6. Why do we group data by region and product category?

Ans: To compare sales performance across different regions and product categories.

Q7. Which charts are suitable for showing sales distribution?

Ans: Bar charts and pie charts.

Q8. What is a stacked bar plot?

Ans: A stacked bar plot shows multiple categories as parts of a whole.

Q9. How do you handle missing or inconsistent region names?

Ans: By removing extra spaces, standardizing text cases, correcting spelling, and handling missing values.

Q10. How does aggregation help business decision-making?

Ans: It helps identify high-performing regions, allocate resources, and improve sales strategies.

Q11. What is the use of sum()?

Ans: It calculates the total of numerical values.

Q12. What is the use of mean()?

Ans: It calculates the average value.

Q13. What is the use of count()?

Ans: It counts the number of non-null records.

Q14. What is the difference between a bar chart and a pie chart?

Ans: A bar chart compares categories, while a pie chart shows their percentage contribution to a whole.

EXPERIMENT 7: TIME SERIES DATA ANALYSIS

Aim

To analyze and visualize stock market data.

Objective

1. To study historical stock prices.
2. To identify trends and patterns.
3. To calculate moving averages.
4. To analyze seasonality and correlation.
5. To forecast future stock prices.

Dataset

Stock_Prices.csv

Theory

1. Time Series Data

Time series data is a sequence of observations recorded at regular time intervals, such as daily stock prices, monthly sales, or hourly temperatures.

In time series analysis, the order of time is important.

2. Stock Market Data Analysis

It involves studying historical and real-time stock data to understand price movements, trends, and patterns.

3. Types of Stock Market Analysis

- Technical Analysis: Uses historical prices, charts, and indicators to identify market trends.
- Fundamental Analysis: Examines a company's financial health, earnings, assets, and liabilities.
- Sentiment Analysis: Studies news, social media, and investor behavior to understand market sentiment.

4. Datetime Format

Datetime format represents dates and times in a format that Python can understand and process.

Example:

df['Date'] = pd.to_datetime(df['Date'])

5. Moving Average

A moving average calculates the average stock price over a specific period. It smooths short-term fluctuations and highlights long-term trends.

Example:

df['MA'] = df['Close'].rolling(window=7).mean()

6. Seasonality

Seasonality refers to repeating patterns that occur at regular intervals, such as weekly, monthly, or yearly.

7. Correlation

Correlation measures the relationship between two variables, such as stock prices and trading volume.

8. ARIMA Model

ARIMA stands for AutoRegressive Integrated Moving Average. It is a statistical model used for time series forecasting.

9. Components of Time Series

- Trend: Long-term upward or downward movement.
- Seasonality: Repeating patterns at regular intervals.
- Cyclic: Long-term fluctuations without a fixed regular period.
- Random Noise: Unpredictable variations.

10. Model Evaluation

Model evaluation checks the accuracy and reliability of predictions.

Common metrics:

- MAE: Mean Absolute Error.
- MSE: Mean Squared Error.
- RMSE: Root Mean Squared Error.
- MAPE: Mean Absolute Percentage Error.

Practical Steps

1. Import Pandas, Matplotlib, and other required libraries.
2. Load Stock_Prices.csv.
3. Explore the dataset.
4. Convert the date column into datetime format.
5. Plot stock prices using line charts.

6. Calculate moving averages.
7. Analyze seasonality.
8. Calculate correlation between stock prices and trading volume.
9. Apply ARIMA or exponential smoothing models for forecasting.
10. Evaluate the forecasting model.

Conclusion

In this experiment, we analyzed historical stock market data using time series analysis. We identified trends, calculated moving averages, studied seasonality and correlation, and explored forecasting techniques.

Viva Questions and Answers

Q1. What is time series data?

Ans: Time series data is a sequence of observations recorded at regular time intervals.

Q2. How is time series data different from normal data?

Ans: In time series data, the order of observations and time dependency are important.

Q3. Why do we convert the date column into datetime format?

Ans: To perform time-based operations such as resampling, rolling averages, and time series plotting.

Q4. Which libraries are commonly used for time series analysis?

Ans: Pandas, Matplotlib, Seaborn, and Statsmodels.

Q5. Which graph is used to visualize stock price trends?

Ans: A line plot.

Q6. What is a moving average?

Ans: A moving average calculates the average value over a specific time period to smooth short-term fluctuations.

Q7. What is the purpose of a rolling average?

Ans: It helps identify underlying trends by reducing short-term fluctuations.

Q8. What is seasonality?

Ans: Seasonality means repeating patterns in data at regular time intervals.

Q9. How can we detect seasonality?

Ans: By plotting data over time, decomposing the time series, or using autocorrelation plots.

Q10. What is correlation in stock market analysis?

Ans: Correlation measures the relationship between stock prices and other variables, such as trading volume.

Q11. What is ARIMA?

Ans: ARIMA stands for AutoRegressive Integrated Moving Average. It is used for time series forecasting.

Q12. What are the four components of time series?

Ans: Trend, seasonality, cyclic patterns, and random noise.

Q13. What is trend?

Ans: Trend is the long-term upward or downward movement in data.

Q14. What is random noise?

Ans: Random noise refers to unpredictable variations in time series data.

Q15. What is forecasting?

Ans: Forecasting is the process of predicting future values using historical data.

Q16. What is exponential smoothing?

Ans: Exponential smoothing is a forecasting technique that gives more weight to recent observations.

Q17. What is MAE?

Ans: MAE stands for Mean Absolute Error. It measures the average absolute difference between actual and predicted values.

Q18. What is RMSE?

Ans: RMSE stands for Root Mean Squared Error. It measures prediction error and gives more weight to larger errors.

Q19. What is a candlestick chart?

Ans: A candlestick chart shows open, high, low, and closing prices for a specific time period.

Q20. What is technical analysis?

Ans: Technical analysis studies historical stock prices and trading volumes to identify possible market movements.

IMPORTANT COMMON DMV VIVA QUESTIONS

Q1. What is Data Modeling?

Ans: Data modeling is the process of creating a structured representation of data and its relationships.

Q2. What is Data Visualization?

Ans: Data visualization is the graphical representation of information using charts and graphs.

Q3. What is EDA?

Ans: EDA stands for Exploratory Data Analysis. It is used to understand datasets through statistics and visualizations.

Q4. What is Python?

Ans: Python is a high-level, interpreted programming language used for data analysis, automation, and machine learning.

Q5. What is Pandas?

Ans: Pandas is a Python library used for data manipulation and analysis.

Q6. What is NumPy?

Ans: NumPy is a Python library used for numerical calculations and array operations.

Q7. What is Seaborn?

Ans: Seaborn is a Python library used for statistical data visualization.

Q8. What is Scikit-learn?

Ans: Scikit-learn is a Python library used for machine learning.

Q9. What is Jupyter Notebook?

Ans: Jupyter Notebook is an interactive environment used to write and execute Python code and display outputs.

Q10. What is a CSV file?

Ans: CSV stands for Comma-Separated Values. It stores tabular data in plain text format.

Q11. What is the difference between Pandas and NumPy?

Ans: Pandas is mainly used for data manipulation and analysis, while NumPy is used for numerical operations and arrays.

Q12. What is the difference between mean, median, and mode?

Ans:

- Mean: Average value.
- Median: Middle value after sorting.
- Mode: Most frequently occurring value.

Q13. What is a histogram?

Ans: A histogram represents the frequency distribution of numerical data.

Q14. What is a heatmap?

Ans: A heatmap represents data using colors to show values and patterns.

Q15. What is the difference between supervised and unsupervised learning?

Ans: Supervised learning uses labeled data, while unsupervised learning uses unlabeled data.

Q16. What is a correlation matrix?

Ans: A correlation matrix shows the correlation between multiple numerical variables.

Q17. What is data preprocessing?

Ans: Data preprocessing is the process of cleaning and transforming raw data before analysis or machine learning.

Q18. What is a library in Python?

Ans: A library is a collection of pre-written code that provides reusable functions and tools.

IMPORTANT PYTHON FUNCTIONS – QUICK REVISION

Function	Purpose
pd.read_csv()	Read CSV file
pd.read_excel()	Read Excel file

pd.read_json()	Read JSON file
df.head()	Display first five rows
df.tail()	Display last five rows
df.info()	Dataset information
df.describe()	Statistical summary
df.shape	Number of rows and columns
df.columns	Display column names
df.isnull().sum()	Count missing values
df.dropna()	Remove missing values
df.fillna()	Fill missing values
df.drop_duplicates()	Remove duplicates
pd.merge()	Merge datasets
df.groupby()	Group data
df.sort_values()	Sort data
pd.to_datetime()	Convert to datetime
plt.plot()	Line chart
plt.bar()	Bar chart
plt.hist()	Histogram
plt.pie()	Pie chart
plt.scatter()	Scatter plot
plt.boxplot()	Box plot
plt.show()	Display graph

END OF DMV EXPERIMENTS 1 TO 7 – VIVA NOTES